

TWINKLE GROVER

Data Scientist | Machine Learning | Fraud & Healthcare Analytics

Senior Business Analyst @ EXL · 8+ Years in Data, ETL & Business Intelligence

+91 7982917278 | kristlin81@gmail.com | linkedin.com/in/twinkle-grover-308385a2 | github.com/kristlingr

PROFESSIONAL SUMMARY

Data Scientist with 8+ years of hands-on data engineering and business analysis experience at EXL. Built four end-to-end ML projects: XGBoost reimbursement cost prediction on 100K+ CMS Medicare records, Random Forest insurance fraud detection with Flask API deployment, real-time n8n fraud pipeline automation, and a SQL Server Data Warehouse with Medallion Architecture. Proficient in Python (XGBoost, Scikit-learn, Pandas), SQL, Tableau, and Power BI. Deep domain expertise in KYC/AML compliance, credit card analytics, ETL validation, and regulatory reporting — enabling ML solutions grounded in real business context.

TECHNICAL SKILLS

ML / AI: XGBoost, Random Forest, Logistic Regression, Scikit-learn, Feature Engineering, EDA, Threshold Optimisation, Class Balancing

Python Stack: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Flask API, Jupyter Notebook

Data Engineering: SQL Server, ETL Pipelines, Medallion Architecture (Bronze/Silver/Gold), Star Schema, Data Warehouse Design, SSMS, PostgreSQL, SAS

Automation & BI: n8n (Webhook/API/Conditional Routing), Tableau, Power BI, Excel (VBA/Macros), Google Sheets

Domains: Healthcare Claims Analytics, Insurance Fraud Detection, KYC/AML, Credit Card Analytics, Revenue Assurance, Data Governance

DATA SCIENCE & ENGINEERING PROJECTS

Healthcare Cost Prediction — CMS Medicare Claims (XGBoost)

Python · XGBoost · Pandas · Scikit-learn · Matplotlib | github.com/kristlingr/CMS-SynPUF-US-Healthcare-Claims-Fraud

- Ingested and processed 100,000+ CMS DE-SynPUF Medicare records, merging beneficiary and carrier claim lines across inpatient, outpatient, and carrier cost dimensions.
- Engineered domain features: TOTAL_CHRONIC (11 chronic conditions), cost aggregates, FULL_YEAR_COVERAGE flag, and beneficiary age; applied log1p transformation on skewed reimbursement target.
- Trained XGBoost (lr=0.05, max_depth=6, 500 rounds with early stopping) — outperforming Linear Regression baseline on MAE, RMSE, and R²; delivered feature importance and Actual vs. Predicted charts.

End-to-End Insurance Fraud Detection System

Python · Scikit-learn · Flask · n8n · Tableau | github.com/kristlingr/Insurance-Fraud-detection-Model-System

- Built full ML pipeline: EDA, feature engineering (claim deviation, claim ratio, policy age), balanced Random Forest with custom threshold (0.2) to maximise fraud recall; risk-segmented claims into High/Medium/Low tiers.
- Deployed model as Flask REST API; orchestrated claim intake via n8n; delivered Tableau dashboards with ROC-AUC curve, feature importance, and fraud trend reporting.

Real-Time Fraud Detection Pipeline — n8n Automation

n8n · Python · Outlook · Google Sheets | github.com/kristlingr/insurance-fraud-detection-n8n

- Designed two-version automated fraud pipeline: v1 (rule-based Python logic) upgraded to v2 (ML API-integrated real-time scoring); webhook ingestion with conditional Outlook alerts and full Google Sheets audit log.

SQL Server Data Warehouse — Medallion Architecture

SQL Server · SSMS · Star Schema · ETL | github.com/kristlingr/SQL-DataWarehouse_project

- Architected modern data warehouse: Bronze (raw CSV ingestion), Silver (cleansing, standardisation), Gold (star schema with fact and dimension tables optimised for analytics).
- Built ETL pipelines and SQL analytics reporting on customer behaviour, product performance, and sales trends; documented data model for business and analytics stakeholders.

PROFESSIONAL EXPERIENCE

Senior Business Analyst — EXL | May 2017 – Present

ETL Validation Framework (Banking & KYC Compliance)

- Engineered SQL ETL validation engine with 4 stored procedures; implemented 40+ automated KYC/CRR rules with full audit trail, error logging, and automated CRRF master table loading.
- Visualised KYC error patterns and data quality trends in Tableau to support real-time compliance team action.

Credit Card Distribution & Revenue Assurance (Citi Cards)

- Designed SQL pipelines for customer segmentation, profiling, and monthly performance reporting; conducted churn analysis identifying data gaps and quality issues.
- Performed large-scale EDA on credit card datasets to detect revenue leakage, incorrect billing, and misaligned transactions; delivered dashboards covering tariff migration, EPDQ, and card alignment.

Financial Analysis, Automation & Regulatory Operations

- Conducted COGS, revenue, and cost-driver analysis against P&L statements (IRMS/CSV Health); designed Excel dashboards with macro-automated CSV ingestion (Marpai).
- Built VBA employee database automation tool with validation and staffing integration; managed 50+ state-level regulatory filings and SLA tracking across insurance domain projects.

EDUCATION

Bachelor of Commerce (B.Com) — University of Delhi | 2016